

MC-210
Differential Equations
Unit-3 Assignment-3

Q.1 Form the partial differential equations by eliminating the arbitrary constants from the followings:

- (a) $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ *Ans* : $2z = px + qy$
- (b) $(x-a)^2 + (y-b)^2 + z^2 = c^2$ *Ans* : $z^2(p^2 + q^2 + 1) = c^2$
- (c) $x^2 + y^2 + (z-c)^2 = a^2$ *Ans* : $yp - xq = 0$
- (d) $x^2 + y^2 = (z-c)^2 \tan^2 \alpha$ *Ans* : $yp - xq = 0$
- (e) $2z = (ax+y)^2 + b$ *Ans* : $z(px+qy) = z^2 - 1$
- (f) $z = xy + y\sqrt{x^2 - a^2} + b$ *Ans* : $px + qy = pq$

Q.2 Form the partial differential equations by eliminating the arbitrary functions from the followings:

- (a) $xyz = \phi(x+y+z)$ *Ans* : $x(y-z)p + y(z-x)q = z(x-y)$
- (b) $z = f(x^2 - y^2)$ *Ans* : $py + qx = 0$
- (c) $f(xy + z^2, x + y + z) = 0$ *Ans* : $p(x - 2z) + q(2z - y) = y - x$
- (d) $f(x^2 + y^2, z - xy) = 0$ *Ans* : $py - qx = y^2 - x^2$
- (e) $lx + my + nz = \phi(x^2 + y^2 + z^2)$ *Ans* : $(mz - ny)p + (nx - lz)q = ly - mx$
- (f) $z = f(x)g(y)$ *Ans* : $z \frac{\partial^2 z}{\partial x \partial y} - \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = 0$

Q.3 Solve the following partial differential equations.

$$(a) \quad y^2p - xyq = x(z - 2y) \quad \text{Ans : } x^2 + y^2 = f(y^2 - yz)$$

$$(b) \quad (p - q)z = z^2 + (x + y)^2 \quad \text{Ans : } \phi(x + y, x - \frac{1}{2}\log(z^2 + (x + y)^2))$$

$$(c) \quad (x^2 - y^2 - z^2)p + 2xyq = 2xz \quad \text{Ans : } \phi(x^2 + y^2 + z^2, \frac{y}{z}) = 0$$

$$(d) \quad x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2) \quad \text{Ans : } \phi(x^2 + y^2 - 2z, xyz) = 0$$

$$(e) \quad x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2) \quad \text{Ans : } \phi(x^2 + y^2 + z^2, \frac{yz}{x}) = 0$$

$$(f) \quad (z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx \quad \text{Ans : } \phi(x^2 + y^2 + z^2, y^2 - 2yz - z^2) = 0$$

Q.4 Find the integral surface of the quasi-linear PDE $x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z$ which contains the straight line $x + y = 0, z = 1$.

$$\text{Ans: } x^2 + y^2 + 2xyz - 2z + 2 = 0$$

Q.5 Find the integral surface of $x^2p + y^2q + z^2 = 0$, which passes through the curves $xy = x + y, z = 1$.

$$\text{Ans: } 3xyz - 2xy - xz - yz = 0$$

Q.6 Find the integral surface of $4yzp + q + 2y = 0$ passing through $y^2 + z^2 = 2, x + z = 1$.

$$\text{Ans: } z^2 + y^2 + z + x = 3.$$

Q.7 Solve the following non-linear PDE.

$$(a) \quad p^3 + q^3 = 27z \quad \text{Ans : } z(z - 2az) = 2(x + ay) + b$$

$$(b) \quad \sqrt{p} + \sqrt{q} = 2x \quad \text{Ans : } z = \frac{(a+2x)^3}{6} + a^2x + b$$

$$(c) \quad q = xyp^2 \quad \text{Ans : } 16ax - (2z - ay^2 - 2b)^2 = 0$$

$$(d) \quad z^2(p^2 + q^2) = x^2 + y^2 \quad \text{Ans : } z^2 = x\sqrt{x^2 + a^2} + y\sqrt{y^2 - a^2} + a^2\log\frac{x + \sqrt{x^2 + a^2}}{y - \sqrt{y^2 - a^2}} + b$$

$$(e) \quad p^2 + q^2 = x + y \quad \text{Ans : } z = \frac{2}{3}(a+x)^{\frac{3}{2}} + \frac{2}{3}(y-a)^{\frac{3}{2}} + b$$

$$(f) \quad a(p^2z + q^2) = 4 \quad \text{Ans : } (a^2z + 1)^2 = (y + ax + c)^2$$

$$(g) \quad z = px + qy + p^2q^2$$

$$Ans : z = ax + by + a^2b^2$$

Q.8 Solve the following non-linear PDE by Charpit's method.

$$(a) \quad 2(z + xp + yq) = yp^2$$

$$Ans : z = \frac{ax}{y^2} - \frac{a^2}{4y^3} + \frac{b}{y}$$

$$(b) \quad (p^2 + q^2)y = qz$$

$$Ans : z^2 = (ax + b)^2 + a^2y^2$$

$$(c) \quad z = p^2x + q^2y$$

$$Ans : \sqrt{(1+a)}z = \sqrt{ax} + \sqrt{y} + b$$

$$(d) \quad pxy + pq + qy = yz$$

$$Ans : \log(z - ax) = y - a \log(a + y) + b$$

$$(e) \quad p = (qy + z)^2$$

$$Ans : yz = ax + 2\sqrt{ay} + b$$